Results

Descriptives

Descriptives

	Gender	BMI_Category	Has_Hypertension	Has_Diabetes	Has_Other_Condition	Had_Surgery	Alcohol_Consumption_yes_no	Is_Smoker	tobacco_consumer	Describe your level of weekly physical activity	Liver_
N	285	285	285	285	285	285	285	285	285	285	

Frequencies

Frequencies of Gender

Gender	Counts	% of Total	Cumulative %
Female	30	10.5%	10.5%
Male	255	89.5%	100.0%

Frequencies of BMI_Category

BMI_Category	Counts	% of Total	Cumulative %
Normal	153	53.7%	53.7%
Obese	22	7.7%	61.4%
Overweight	97	34.0%	95.4%
Underweight	13	4.6%	100.0%

Frequencies of Has_Hypertension

Has_Hypertension	Counts	% of Total	Cumulative %
No	271	95.1%	95.1%
Yes	14	4.9%	100.0%

Frequencies of Has_Diabetes

Has_Diabetes	Counts	% of Total	Cumulative %
No	273	95.8%	95.8%
Yes	12	4.2%	100.0%

Frequencies of Has_Other_Condition

Has_Other_Condition	Counts	% of Total	Cumulative %
No	274	96.1%	96.1%
Yes	11	3.9%	100.0%

Frequencies of Had_Surgery

Had_Surgery	Counts	% of Total	Cumulative %
No	250	87.7%	87.7%
Yes	35	12.3%	100.0%

Frequencies of Alcohol_Consumption_yes_no

Alcohol_Consumption_yes_no	Counts	% of Total	Cumulative %
No	214	75.1%	75.1%
Yes	71	24.9%	100.0%

Frequencies of Is_Smoker

Is_Smoker	Counts	% of Total	Cumulative %
No	260	91.2%	91.2%
Yes	25	8.8%	100.0%

Frequencies of tobacco_consumer

tobacco_consumer	Counts	% of Total	Cumulative %
No	280	98.2%	98.2%
Yes	5	1.8%	100.0%

Frequencies of Describe your level of weekly physical activity

Describe your level of weekly physical activity	Counts	% of Total	Cumulative %
Active	49	17.2%	17.2%
Light	91	31.9%	49.1%
Moderate	129	45.3%	94.4%
Sedentary	16	5.6%	100.0%

Frequencies of Liver_Status_Label

Liver_Status_Label	Counts	% of Total	Cumulative %
ALD	33	11.6%	11.6%
Hyperbilirubinemia	13	4.6%	16.1%
NAFLD	41	14.4%	30.5%
Normal	169	59.3%	89.8%
Others	29	10.2%	100.0%

Frequencies of Normal_Abnormal_Liver

Normal_Abnormal_Liver	Counts	% of Total	Cumulative %
Abnormal	116	40.7%	40.7%
Normal	169	59.3%	100.0%

Frequencies of With_underreported_Alcohol_History

With_underreported_Alcohol_History	Counts	% of Total	Cumulative %
ALD (Confirmed)	7	2.5%	2.5%
ALD (Probable)	26	9.1%	11.6%
Hyperbilirubinemia	13	4.6%	16.1%
NAFLD	41	14.4%	30.5%
Normal	169	59.3%	89.8%
Others	29	10.2%	100.0%

Frequencies of Future_Risk_Category

Future_Risk_Category	Counts	% of Total	Cumulative %
High Risk	4	1.4%	1.4%
Low Risk	212	74.4%	75.8%
Moderate Risk	69	24.2%	100.0%

Contingency Tables (Across Gender)

Contingency Tables

Gender		Liver_Status_Label					Total
		ALD	Hyperbilirubinemia	NAFLD	Normal	Others	
Female	Observed	2	1	3	24	0	30
	% within row	6.7%	3.3%	10.0%	80.0%	0.0%	100.0%
Male	Observed	31	12	38	145	29	255
	% within row	12.2%	4.7%	14.9%	56.9%	11.4%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

χ^2 Tests

	Value
N	285

Contingency Tables (Across Age Group)

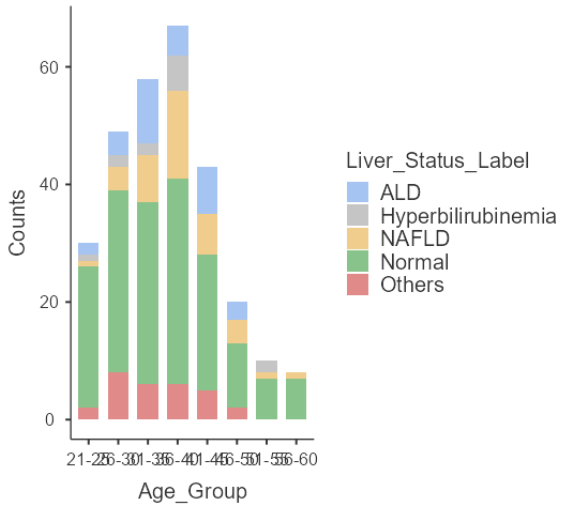
Contingency Tables

Age_Group		Liver_Status_Label					Total
		ALD	Hyperbilirubinemia	NAFLD	Normal	Others	
21-25	Observed	2	1	1	24	2	30
	% within row	6.7%	3.3%	3.3%	80.0%	6.7%	100.0%
26-30	Observed	4	2	4	31	8	49
	% within row	8.2%	4.1%	8.2%	63.3%	16.3%	100.0%
31-35	Observed	11	2	8	31	6	58
	% within row	19.0%	3.4%	13.8%	53.4%	10.3%	100.0%
36-40	Observed	5	6	15	35	6	67
	% within row	7.5%	9.0%	22.4%	52.2%	9.0%	100.0%
41-45	Observed	8	0	7	23	5	43
	% within row	18.6%	0.0%	16.3%	53.5%	11.6%	100.0%
46-50	Observed	3	0	4	11	2	20
	% within row	15.0%	0.0%	20.0%	55.0%	10.0%	100.0%
51-55	Observed	0	2	1	7	0	10
	% within row	0.0%	20.0%	10.0%	70.0%	0.0%	100.0%
56-60	Observed	0	0	1	7	0	8
	% within row	0.0%	0.0%	12.5%	87.5%	0.0%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

χ^2 Tests

	Value
N	285

Plots



Descriptives

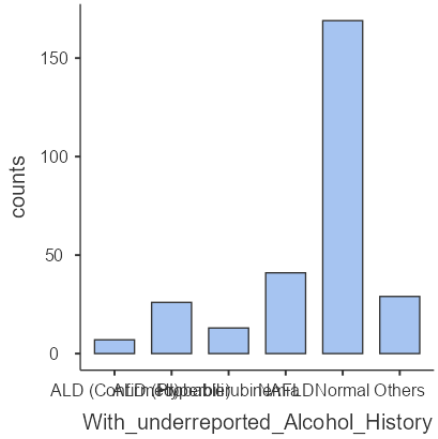
Frequencies

Frequencies of With_underreported_Alcohol_History

With_underreported_Alcohol_History	Counts	% of Total	Cumulative %
ALD (Confirmed)	7	2.5%	2.5%
ALD (Probable)	26	9.1%	11.6%
Hyperbilirubinemia	13	4.6%	16.1%
NAFLD	41	14.4%	30.5%
Normal	169	59.3%	89.8%
Others	29	10.2%	100.0%

Plots

With_underreported_Alcohol_History



Descriptives

Descriptives

	Age	work_experience_yrs	Height_cm	Weight_kg	BMI	ALT	AST	Bilirubin_Total	Bilirubin_Direct	Bilirubin_Indirect	AST_ALT_Ratio	Future_Risk_S
N	285	285	285	285	285	285	285	285	285	285	285	2
Mean	36.2	11.7	167	69.5	24.8	32.7	44.3	0.594	0.116	0.478	1.54	1
Median	36	10	167	69.0	24.7	29.4	37.5	0.510	0.100	0.420	1.30	
Standard deviation	8.58	7.72	7.93	11.8	3.62	15.7	27.7	0.339	0.0826	0.292	0.957	9.
IQR	12.0	12.0	10.0	14.0	4.12	16.9	11.0	0.380	0.0600	0.320	0.772	1
Minimum	21	0	140	35.0	14.4	7.36	15.7	0.100	0.0100	0.0100	0.269	
Maximum	65	34	188	125	38.9	148	216	1.90	0.900	1.56	7.20	
Skewness	0.430	0.588	-0.0595	0.462	0.411	2.20	3.63	1.35	3.76	1.27	2.89	0.9
Std. error skewness	0.144	0.144	0.144	0.144	0.144	0.144	0.144	0.144	0.144	0.144	0.144	0.1
Kurtosis	0.0597	-0.156	0.583	1.79	1.49	10.5	14.9	1.86	28.7	1.66	11.0	0.2
Std. error kurtosis	0.288	0.288	0.288	0.288	0.288	0.288	0.288	0.288	0.288	0.288	0.288	0.2
Shapiro-Wilk W	0.981	0.954	0.987	0.973	0.975	0.856	0.572	0.890	0.748	0.904	0.727	0.9
Shapiro-Wilk p	<.001	<.001	0.014	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.0

"Descriptive statistics were computed for all continuous variables (N=285). Normality was assessed using the Shapiro-Wilk test. All continuous variables showed significant deviation from normality ($p < 0.05$ for all), including Age ($W=0.981$, $p<0.001$), BMI ($W=0.975$, $p<0.001$), ALT ($W=0.856$, $p<0.001$), AST ($W=0.572$, $p<0.001$), Bilirubin Total ($W=0.890$, $p<0.001$), and AST:ALT Ratio ($W=0.727$, $p<0.001$). This was further confirmed by high skewness values, particularly for AST (skewness=3.63), Bilirubin Direct (3.76), and AST:ALT Ratio (2.89), indicating strongly right-skewed distributions. Accordingly, data are expressed as Median [IQR] and non-parametric tests were used for all subsequent comparisons."

Independent Samples T-Test

Independent Samples T-Test

		Statistic	p		Effect Size
ALT	**Mann-Whitney U**	6580	<.001	Rank biserial correlation	-0.3288
AST	**Mann-Whitney U**	1248	<.001	Rank biserial correlation	-0.8727
Bilirubin_Total	**Mann-Whitney U**	8950	0.213	Rank biserial correlation	-0.0869
Bilirubin_Direct	**Mann-Whitney U**	9493	0.651	Rank biserial correlation	-0.0316
Bilirubin_Indirect	**Mann-Whitney U**	8865	0.170	Rank biserial correlation	-0.0956
AST_ALT_Ratio	**Mann-Whitney U**	7285	<.001	Rank biserial correlation	-0.2568
Age	**Mann-Whitney U**	8958	0.217	Rank biserial correlation	-0.0861
BMI	**Mann-Whitney U**	8614	0.082	Rank biserial correlation	-0.1212
Future_Risk_Score	**Mann-Whitney U**	5538	<.001	Rank biserial correlation	-0.4350

Note. H_a $\mu_{Abnormal} \neq \mu_{Normal}$

Group Descriptives

	Group	N	Mean	Median	SD	SE
ALT	**Abnormal**	116	38.940	34.200	19.579	1.81789
	Normal	169	28.356	27.700	10.4568	0.80437
AST	**Abnormal**	116	60.832	46.900	37.236	3.45725
	Normal	169	33.013	35.200	5.7073	0.43902
Bilirubin_Total	**Abnormal**	116	0.648	0.505	0.405	0.03763
	Normal	169	0.556	0.510	0.2798	0.02152
Bilirubin_Direct	**Abnormal**	116	0.126	0.100	0.107	0.00990
	Normal	169	0.109	0.100	0.0603	0.00464
Bilirubin_Indirect	**Abnormal**	116	0.523	0.420	0.347	0.03226
	Normal	169	0.447	0.410	0.2441	0.01878
AST_ALT_Ratio	**Abnormal**	116	1.867	1.544	1.320	0.12258
	Normal	169	1.308	1.190	0.4756	0.03659
Age	**Abnormal**	116	36.741	36.000	7.503	0.69665
	Normal	169	35.822	35.000	9.2565	0.71204
BMI	**Abnormal**	116	25.243	24.740	3.829	0.35555
	Normal	169	24.492	24.465	3.4555	0.26581
Future_Risk_Score	**Abnormal**	116	19.155	17.500	8.996	0.83528
	Normal	169	12.959	11.000	8.8313	0.67933

One-Way ANOVA (Non-parametric)

Kruskal-Wallis

	χ^2	df	p	ε^2
Age	4.50	4	0.343	0.0158
BMI	30.93	4	<.001	0.1089
AST	181.57	4	<.001	0.6393
ALT	53.84	4	<.001	0.1896
Bilirubin_Total	38.15	4	<.001	0.1343
Bilirubin_Direct	40.10	4	<.001	0.1412
Bilirubin_Indirect	31.57	4	<.001	0.1112
AST_ALT_Ratio	102.85	4	<.001	0.3622
Future_Risk_Score	61.82	4	<.001	0.2177

Dwass-Steel-Critchlow-Fligner pairwise comparisons

Pairwise comparisons - Age

		W	p
ALD	**Hyperbilirubinemia**	-0.173	1.000
ALD	**NAFLD**	1.270	0.898
ALD	**Normal**	-1.032	0.950
ALD	**Others**	-1.318	0.885
Hyperbilirubinemia	**NAFLD**	1.464	0.839
Hyperbilirubinemia	**Normal**	-0.371	0.999
Hyperbilirubinemia	**Others**	-0.578	0.994
NAFLD	**Normal**	-2.679	0.320
NAFLD	**Others**	-2.449	0.415
Normal	**Others**	-0.439	0.998

Pairwise comparisons - BMI

		W	p
ALD	**Hyperbilirubinemia**	-2.520	0.384
ALD	**NAFLD**	4.937	0.004
ALD	**Normal**	-0.223	1.000
ALD	**Others**	-0.938	0.964
Hyperbilirubinemia	**NAFLD**	5.796	<.001
Hyperbilirubinemia	**Normal**	2.539	0.376
Hyperbilirubinemia	**Others**	1.637	0.776
NAFLD	**Normal**	-6.778	<.001
NAFLD	**Others**	-5.894	<.001
Normal	**Others**	-1.015	0.953

Pairwise comparisons - AST

		W	p
ALD	**Hyperbilirubinemia**	-6.35	<.001
ALD	**NAFLD**	-9.14	<.001
ALD	**Normal**	-12.85	<.001
ALD	**Others**	-7.36	<.001
Hyperbilirubinemia	**NAFLD**	4.27	0.022
Hyperbilirubinemia	**Normal**	-3.06	0.195
Hyperbilirubinemia	**Others**	5.68	<.001
NAFLD	**Normal**	-11.84	<.001
NAFLD	**Others**	3.25	0.144
Normal	**Others**	12.17	<.001

Pairwise comparisons - ALT

		W	p
ALD	**Hyperbilirubinemia**	-0.190	1.000
ALD	**NAFLD**	6.867	<.001
ALD	**Normal**	-0.341	0.999
ALD	**Others**	3.202	0.157
Hyperbilirubinemia	**NAFLD**	5.351	0.001
Hyperbilirubinemia	**Normal**	0.332	0.999
Hyperbilirubinemia	**Others**	3.155	0.168
NAFLD	**Normal**	-9.663	<.001
NAFLD	**Others**	-5.075	0.003
Normal	**Others**	4.001	0.038

Pairwise comparisons - Bilirubin_Total

		W	p
ALD	**Hyperbilirubinemia**	6.850	<.001
ALD	**NAFLD**	0.415	0.998
ALD	**Normal**	-0.214	1.000
ALD	**Others**	-2.156	0.547
Hyperbilirubinemia	**NAFLD**	-7.112	<.001
Hyperbilirubinemia	**Normal**	-8.009	<.001
Hyperbilirubinemia	**Others**	-7.264	<.001
NAFLD	**Normal**	-1.232	0.908
NAFLD	**Others**	-3.561	0.087
Normal	**Others**	-2.655	0.330

Pairwise comparisons - Bilirubin_Direct

		W	p
ALD	**Hyperbilirubinemia**	6.8598	<.001
ALD	**NAFLD**	0.0540	1.000
ALD	**Normal**	-0.2146	1.000
ALD	**Others**	-3.2737	0.140
Hyperbilirubinemia	**NAFLD**	-7.1932	<.001
Hyperbilirubinemia	**Normal**	-7.7773	<.001
Hyperbilirubinemia	**Others**	-7.1905	<.001
NAFLD	**Normal**	-0.2011	1.000
NAFLD	**Others**	-3.6725	0.071
Normal	**Others**	-3.9344	0.043

Pairwise comparisons - Bilirubin_Indirect

		W	p
ALD	**Hyperbilirubinemia**	6.057	<.001
ALD	**NAFLD**	0.461	0.998
ALD	**Normal**	-0.124	1.000
ALD	**Others**	-1.507	0.824
Hyperbilirubinemia	**NAFLD**	-6.340	<.001
Hyperbilirubinemia	**Normal**	-7.395	<.001
Hyperbilirubinemia	**Others**	-6.949	<.001
NAFLD	**Normal**	-1.491	0.830
NAFLD	**Others**	-2.961	0.223
Normal	**Others**	-2.057	0.592

Pairwise comparisons - AST_ALT_Ratio

		W	p
ALD	**Hyperbilirubinemia**	-7.400	<.001
ALD	**NAFLD**	-10.404	<.001
ALD	**Normal**	-12.157	<.001
ALD	**Others**	-9.547	<.001
Hyperbilirubinemia	**NAFLD**	-4.421	0.015
Hyperbilirubinemia	**Normal**	-2.863	0.254
Hyperbilirubinemia	**Others**	0.250	1.000
NAFLD	**Normal**	5.302	0.002
NAFLD	**Others**	5.809	<.001
Normal	**Others**	4.040	0.035

Pairwise comparisons - Future_Risk_Score

		W	p
ALD	**Hyperbilirubinemia**	-4.513	0.012
ALD	**NAFLD**	3.079	0.188
ALD	**Normal**	-6.058	<.001
ALD	**Others**	-2.240	0.508
Hyperbilirubinemia	**NAFLD**	6.246	<.001
Hyperbilirubinemia	**Normal**	0.720	0.987
Hyperbilirubinemia	**Others**	3.073	0.190
NAFLD	**Normal**	-9.551	<.001
NAFLD	**Others**	-5.101	0.003
Normal	**Others**	3.421	0.110

One-Way ANOVA (Non-parametric)

Kruskal-Wallis

	χ^2	df	p	ε^2
Age	14.868	1	<.001	0.05235
BMI	3.862	1	0.049	0.01360
ALT	25.354	1	<.001	0.08927
AST	24.152	1	<.001	0.08504
Bilirubin_Total	5.832	1	0.016	0.02053
Bilirubin_Direct	0.411	1	0.522	0.00145
Bilirubin_Indirect	6.329	1	0.012	0.02228
AST_ALT_Ratio	2.611	1	0.106	0.00919
Future_Risk_Score	11.065	1	<.001	0.03896

Dwass-Steel-Critchlow-Fligner pairwise comparisons

Pairwise comparisons - Age

		W	p
Female	**Male**	5.45	<.001

Pairwise comparisons - BMI

		W	p
Female	**Male**	2.78	0.049

Pairwise comparisons - ALT

		W	p
Female	**Male**	7.12	<.001

Pairwise comparisons - AST

		W	p
Female	**Male**	6.95	<.001

Pairwise comparisons - Bilirubin_Total

		W	p
Female	**Male**	3.42	0.016

Pairwise comparisons - Bilirubin_Direct

		W	p
Female	**Male**	0.906	0.522

Pairwise comparisons - Bilirubin_Indirect

		W	p
Female	**Male**	3.56	0.012

Pairwise comparisons - AST_ALT_Ratio

		W	p
Female	**Male**	-2.29	0.106

Pairwise comparisons - Future_Risk_Score

		W	p
Female	**Male**	4.70	<.001

Contingency Tables

Contingency Tables

	.		
.	.	.	Total
.	.	.	.
.	.	.	.
Total	.	.	.

χ^2 Tests

	Value
N	.

Paired Samples Contingency Tables

Contingency Tables

	.		
.	.	.	Total
.	.	.	.
.	.	.	.
Total	.	.	.

McNemar Test

	Value	df	p
χ^2	.	.	.
N	.		

Contingency Tables

Contingency Tables

		Liver_Status_Label					
Alcohol_Consumption_yes_no		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
No	Observed	26	12	29	130	17	214
	% within row	12.1%	5.6%	13.6%	60.7%	7.9%	100.0%
Yes	Observed	7	1	12	39	12	71
	% within row	9.9%	1.4%	16.9%	54.9%	16.9%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

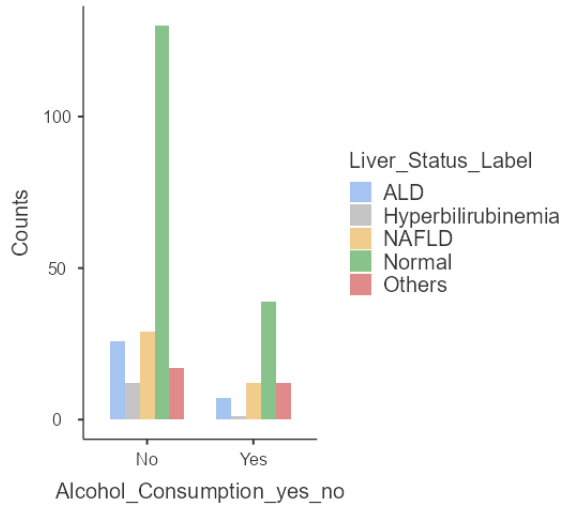
χ^2 Tests

	Value	df	p
χ^2	7.23	4	0.124
Fisher's exact test			0.143
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.159

Plots



Contingency Tables

Contingency Tables

		Liver_Status_Label					
Is_Smoker		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
No	Observed	28	13	36	158	25	260
	% within row	10.8%	5.0%	13.8%	60.8%	9.6%	100.0%
Yes	Observed	5	0	5	11	4	25
	% within row	20.0%	0.0%	20.0%	44.0%	16.0%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

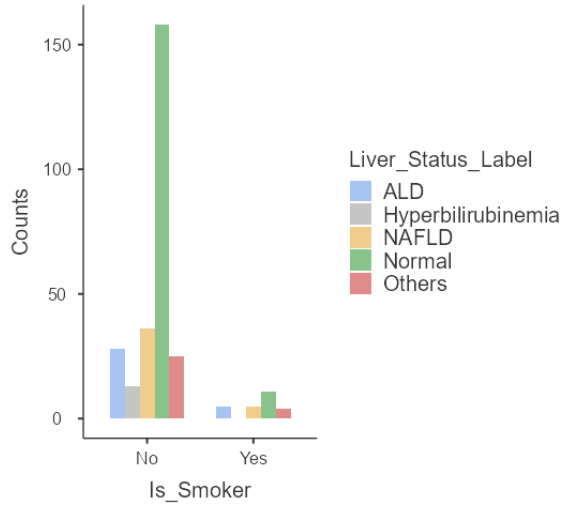
χ^2 Tests

	Value	**df**	**p**
χ^2	5.52	4	0.238
Fisher's exact test			0.209
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.139

Plots



Contingency Tables

Contingency Tables

		Liver_Status_Label					
tobacco_consumer		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
No	Observed	33	13	39	166	29	280
	% within row	11.8%	4.6%	13.9%	59.3%	10.4%	100.0%
Yes	Observed	0	0	2	3	0	5
	% within row	0.0%	0.0%	40.0%	60.0%	0.0%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

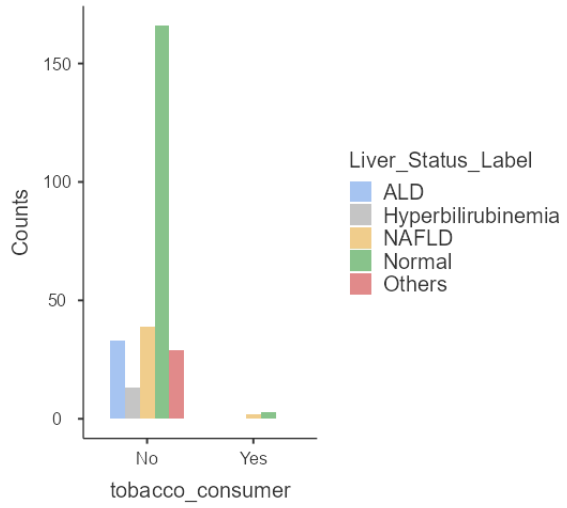
χ^2 Tests

	Value	df	p
χ^2	3.66	4	0.454
Fisher's exact test			0.522
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.113

Plots



Contingency Tables

Contingency Tables

		Liver_Status_Label					
Has_Hypertension		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
No	Observed	32	13	38	160	28	271
	% within row	11.8%	4.8%	14.0%	59.0%	10.3%	100.0%
Yes	Observed	1	0	3	9	1	14
	% within row	7.1%	0.0%	21.4%	64.3%	7.1%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

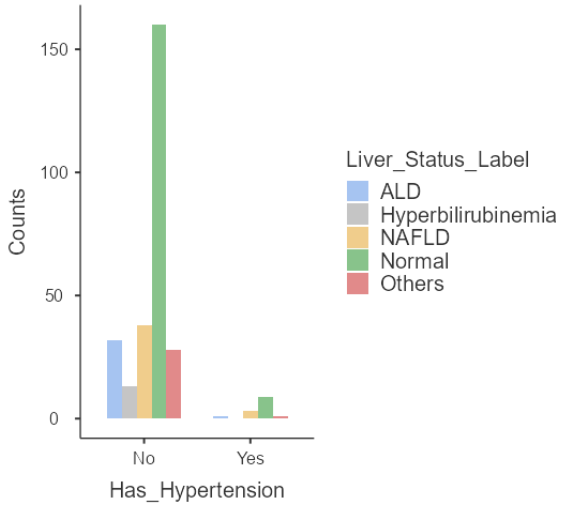
χ^2 Tests

	Value	df	p
χ^2	1.62	4	0.804
Fisher's exact test			0.953
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.0755

Plots



Contingency Tables

Contingency Tables

		Liver_Status_Label					
Has_Diabetes		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
No	Observed	33	13	38	160	29	273
	% within row	12.1%	4.8%	13.9%	58.6%	10.6%	100.0%
Yes	Observed	0	0	3	9	0	12
	% within row	0.0%	0.0%	25.0%	75.0%	0.0%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

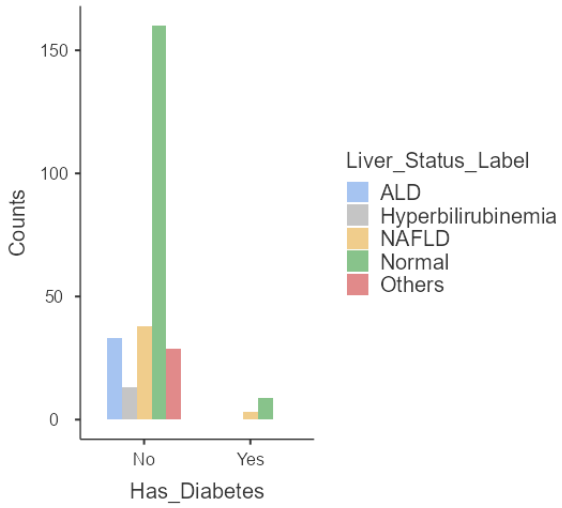
χ^2 Tests

	Value	df	p
χ^2	4.80	4	0.309
Fisher's exact test			0.436
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.130

Plots



Contingency Tables

Contingency Tables

		Liver_Status_Label					
Gender		**ALD**	**Hyperbilirubinemia**	**NAFLD**	**Normal**	**Others**	**Total**
Female	Observed	2	1	3	24	0	30
	% within row	6.7%	3.3%	10.0%	80.0%	0.0%	100.0%
Male	Observed	31	12	38	145	29	255
	% within row	12.2%	4.7%	14.9%	56.9%	11.4%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

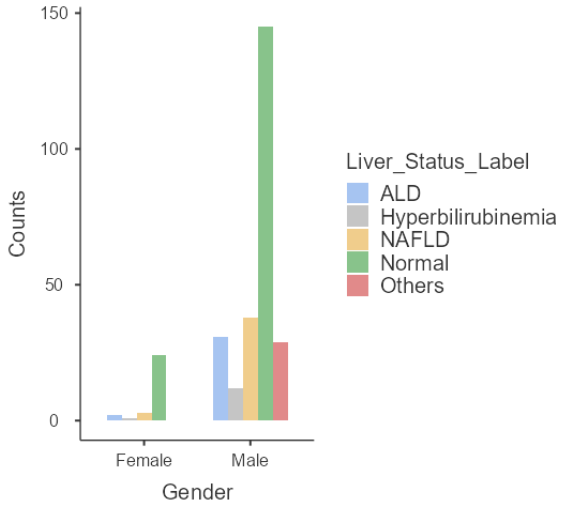
χ^2 Tests

	Value	df	p
χ^2	7.09	4	0.131
Fisher's exact test			0.128
N	285		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.158

Plots



Contingency Tables

Contingency Tables

BMI_Category		Liver_Status_Label					Total
		ALD	Hyperbilirubinemia	NAFLD	Normal	Others	
Normal	Observed	19	11	8	98	17	153
	% within row	12.4%	7.2%	5.2%	64.1%	11.1%	100.0%
Obese	Observed	2	0	10	9	1	22
	% within row	9.1%	0.0%	45.5%	40.9%	4.5%	100.0%
Overweight	Observed	12	1	22	54	8	97
	% within row	12.4%	1.0%	22.7%	55.7%	8.2%	100.0%
Underweight	Observed	0	1	1	8	3	13
	% within row	0.0%	7.7%	7.7%	61.5%	23.1%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

χ^2 Tests

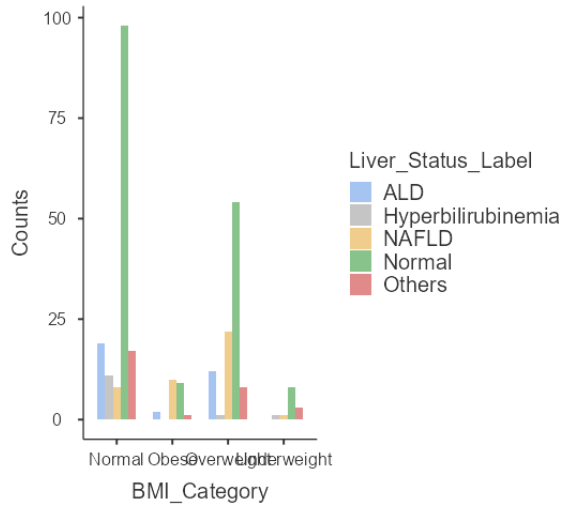
	Value	df	p
χ^2	42.1	12	<.001
Fisher's exact test			<.001[a]
N	285		

[a] Monte Carlo simulation

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.222

Plots



Contingency Tables

Contingency Tables

Describe your level of weekly physical activity		Liver_Status_Label					
		ALD	Hyperbilirubinemia	NAFLD	Normal	Others	Total
Active	Observed	5	2	6	32	4	49
	% within row	10.2%	4.1%	12.2%	65.3%	8.2%	100.0%
Light	Observed	12	4	11	54	10	91
	% within row	13.2%	4.4%	12.1%	59.3%	11.0%	100.0%
Moderate	Observed	13	7	21	73	15	129
	% within row	10.1%	5.4%	16.3%	56.6%	11.6%	100.0%
Sedentary	Observed	3	0	3	10	0	16
	% within row	18.8%	0.0%	18.8%	62.5%	0.0%	100.0%
Total	Observed	33	13	41	169	29	285
	% within row	11.6%	4.6%	14.4%	59.3%	10.2%	100.0%

χ^2 Tests

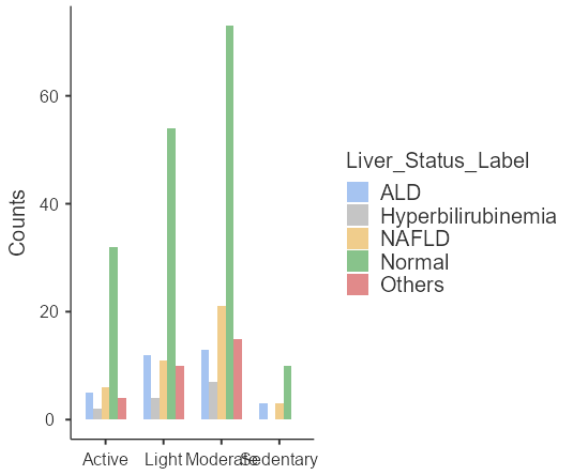
	Value	df	p
χ^2	5.88	12	0.922
Fisher's exact test			0.953[a]
N	285		

[a] Monte Carlo simulation

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.0829

Plots



)escribe your level of weekly physical activity

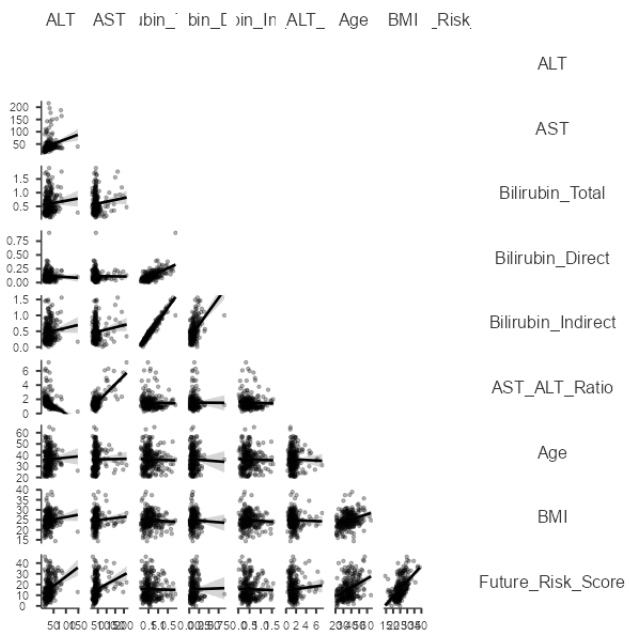
Correlation Matrix

Correlation Matrix

		ALT	AST	Bilirubin_Total	Bilirubin_Direct	Bilirubin_Indirect	AST_ALT_Ratio	Age	BMI	Future_Risk_Score
ALT	**Spearman's rho**	—								
	df	—								
	p-value	—								
AST	**Spearman's rho**	0.354***	—							
	df	283	—							
	p-value	<.001	—							
Bilirubin_Total	**Spearman's rho**	0.058	0.004	—						
	df	283	283	—						
	p-value	0.331	0.951	—						
Bilirubin_Direct	**Spearman's rho**	-0.058	-0.105	0.603***	—					
	df	283	283	283	—					
	p-value	0.327	0.078	<.001	—					
Bilirubin_Indirect	**Spearman's rho**	0.082	0.029	0.981***	0.458***	—				
	df	283	283	283	283	—				
	p-value	0.166	0.628	<.001	<.001	—				
AST_ALT_Ratio	**Spearman's rho**	-0.688***	0.323***	-0.058	-0.026	-0.065	—			
	df	283	283	283	283	283	—			
	p-value	<.001	<.001	0.329	0.658	0.274	—			
Age	**Spearman's rho**	0.105	0.122*	0.017	-0.027	0.006	-0.025	—		
	df	283	283	283	283	283	283	—		
	p-value	0.076	0.040	0.779	0.652	0.926	0.670	—		
BMI	**Spearman's rho**	0.169**	0.121*	0.048	0.044	0.032	-0.116*	0.306***	—	
	df	283	283	283	283	283	283	283	—	
	p-value	0.004	0.042	0.422	0.460	0.592	0.050	<.001	—	
Future_Risk_Score	**Spearman's rho**	0.347***	0.373***	0.029	0.073	0.008	-0.069	0.379***	0.571***	—
	df	283	283	283	283	283	283	283	283	—
	p-value	<.001	<.001	0.626	0.219	0.894	0.247	<.001	<.001	—

Note. * p < .05, ** p < .01, *** p < .001

Plot



Binomial Logistic Regression

Model Fit Measures

			Overall Model Test		
Model	**AIC**	**R^2_N**	**χ^2**	**df**	**p**
1	137	0.856	287	18	<.001

Note. Models estimated using sample size of N=285

Model Coefficients - Liver_Status

						95% Confidence Interval	
Predictor	**Estimate**	**SE**	**Z**	**p**	**Odds ratio**	**Lower**	**Upper**
Intercept	-39.9436	7.7170	-5.1760	<.001	4.50e-18	1.21e -24	1.67e -11
Age	-0.0614	0.0394	-1.5597	0.119	0.940	0.87052	1.02
BMI	0.0380	0.1133	0.3352	0.737	1.039	0.83180	1.30
Gender:							
Male – Female	0.5934	2.7679	0.2144	0.830	1.810	0.00797	410.90
Alcohol_Consumption_yes_no:							
Yes – No	-1.0529	0.7240	-1.4542	0.146	0.349	0.08442	1.44
Is_Smoker:							
Yes – No	2.2799	1.2662	1.8006	0.072	9.775	0.81724	116.93
Has_Hypertension:							
Yes – No	-0.9301	1.0530	-0.8833	0.377	0.395	0.05009	3.11
Has_Diabetes:							
Yes – No	1.3722	1.4778	0.9286	0.353	3.944	0.21780	71.42
Describe your level of weekly physical activity:							
Light – Active	0.4812	0.8261	0.5825	0.560	1.618	0.32049	8.17
Moderate – Active	0.9782	0.8398	1.1649	0.244	2.660	0.51290	13.79
Sedentary – Active	-0.8573	2.6540	-0.3230	0.747	0.424	0.00234	77.05
ALT	0.1904	0.0431	4.4187	<.001	1.210	1.11176	1.32
AST	0.6305	0.1134	5.5617	<.001	1.878	1.50423	2.35
Bilirubin_Total	-12.8901	151.1530	-0.0853	0.932	2.52e -6	5.50e-135	1.16e+123
Bilirubin_Direct	36.8544	151.0769	0.2439	0.807	1.01e+16	2.56e-113	4.00e+144
Bilirubin_Indirect	13.3620	151.1485	0.0884	0.930	635420.711	1.40e-123	2.89e+134
AST_ALT_Ratio	3.4454	1.0659	3.2325	0.001	31.356	3.88200	253.28
Waist_Size_in	0.0292	0.1332	0.2190	0.827	1.030	0.79297	1.34
dietary_preferences_veg_nonveg:							
Vegitarian – Non Vegitarian	-0.0273	0.5998	-0.0456	0.964	0.973	0.30031	3.15

Note. Estimates represent the log odds of "Liver_Status = 1" vs. "Liver_Status = 0"

Prediction

Classification Table – ...

	Predicted		
Observed	**0**	**1**	**% Correct**
0	162	7	95.9
1	12	104	89.7

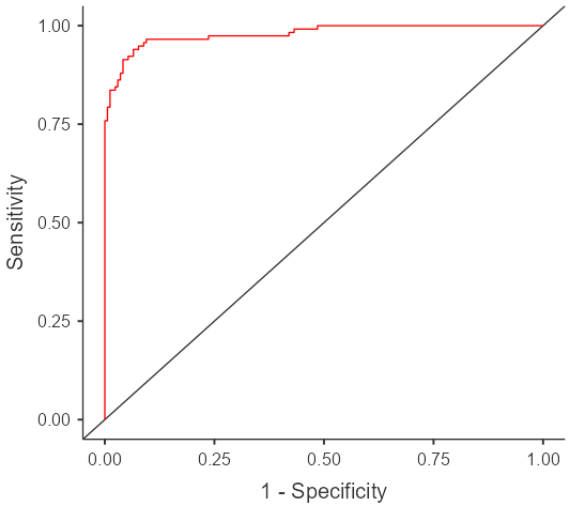
Note. The cut-off value is set to 0.5

Predictive Measures

Accuracy	AUC
0.933	0.979

Note. The cut-off value is set to 0.5

ROC Curve



[4]

References

[1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2024-08-07).

[3] Kerby, D. S. (2014). The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology, 3*, 2165–2228.

[4] Sing, T., Sander, O., Beerenwinkel, N., & Lengauer, T., Unterthiner, T., & Ernst, F. G. M. (2020). *ROCR: Visualizing the Performance of Scoring Classifiers*. [R package]. Retrieved from https://cran.r-project.org/package=ROCR.